RAGURAJ RAJENDRAN

EMBEDDED SOFTWARE ENGINEER | 8+ YEARS EXP.

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 8+ years of experience in embedded systems, robotics software, and IoT solutions. Expertise in firmware architecture, Bluetooth Low Energy (BLE), power optimization, and secure firmware updates. Proficient in ESP32, STM32, and nRF52 microcontrollers, with a strong background in FreeRTOS, Zephyr RTOS, and secure embedded development. Recent experience developing ROS-based robotics software, drivetrain (motor and encoder) control, and Linux/Catkin build systems at a Japanese robotics company. Adept at debugging, troubleshooting, and optimizing low-power battery-operated IoT edge devices and robotics platforms for scalability and reliability. Experienced in working with UK-based and Japanese product clients and contributing to CE-certified products.

TECHNICAL SKILLS

- **Embedded Systems:** ESP32, STM32F4, nRF52840, Atmel Microcontrollers, Nordic SoCs, MCU-based Motor & Encoder Control (Drivetrain)
- **Programming:** C, Python, Embedded C++, Linux Shell Scripting
- **RTOS & Firmware:** FreeRTOS, Zephyr RTOS, Bare-Metal Programming
- **Robotics & Build Systems:** ROS (Robot Operating System), Catkin Build System, STM32-Linux Build System
- **Communication Protocols:** SPI, I2C, UART, CAN, BLE, MQTT(S), MODBUS, HTTPS, SFTP, LoRaWAN, RS485
- **Security:** Secure Boot, Flash Encryption, AES, SHA256, TLS, PKI-based Authentication
- **Debugging Tools:** JTAG/SWD Debugger, Logic Analyzer, Oscilloscope, Wireshark, OpenOCD
- **Development Tools:** STM32CubeIDE, ESP-IDF, Segger Embedded Studio, Atmel Studio, Eclipse, Keil, VS Code
- **Version Control & CI/CD:** Git, Gerrit, Bitbucket

WORK EXPERIENCE

Embedded Software Engineer | Rapyuta Robotics, Chennai (Japanese Robotics Company) — Sep 2025 – Present

- Serve as **component owner** for embedded robotics software modules, driving new feature development, enhancements, and performance optimization.
- Own **end-to-end software responsibilities** from design through deployment, maintaining structured software change tracking and records.
- Collaborate closely with **cross-functional hardware, robotics, and firmware teams** to deliver integrated robotics solutions.
- Develop and maintain **drivetrain control software**, including motor and encoder interfacing, for robotics platforms.
- Build and maintain software using **ROS, Linux, and the Catkin build system**, alongside **STM32-Linux based build pipelines**.
- Work with **Zephyr RTOS and MCU firmware** for embedded robotics components.

Senior Firmware Engineer | NGP Websmart Pvt Ltd, Chennai — March 2022 – Aug 2025

- Designed and developed firmware for **Smart Energy Meter, Energy Saver for 3ph Motor, LoRa/MODBUS Gateways, and Environmental Sensor nodes.**
- Developed firmware for **ESP32-S3, STM32F4, and nRF52840,** ensuring **ultra-low-power** consumption with optimized deep sleep cycles.
- Built a **remote IoT device management platform**, allowing **real-time device health monitoring** and firmware updates.
- **Implemented secure OTA updates**, flash encryption, and **TLS-based secure communication.**
- Developed **custom device drivers** for I2C, SPI, UART, and MODBUS interfaces.
- Integrated **coredump mechanisms** to record crashes in deployed devices, enabling efficient monitoring and debugging.
- **Designed a fault-tolerant state machine** for GSM-based remote connectivity, reducing downtime by **30%.**
- Designed and implemented an **embedded control system to optimize energy consumption in 3-phase induction motors** using thyristor-based phase angle control; developed using **STM32 and STM32CubeIDE**, with real-time power monitoring and adaptive load control features.
- Designed a **custom bootloader for the STM32F411 series** controller for secure OTA firmware updates.

Embedded Software Engineer | Kalycito Infotech Pvt Ltd, Coimbatore (Worked in Robert Bosch, Coimbatore)
Aug 2021 – Feb 2022

- **Developed and executed SET and KUNIT test cases** for existing driver files, ensuring robust functionality.
- **Utilized static code analysis tools** to identify and resolve warnings and bugs, improving code quality.

Embedded Engineer | Rajmall Inventives, Chennai **Jan 2020 – Aug 2021**

- Designed and implemented **BLE 5.0 and LoRa** communication protocols for **real-time nurse call systems**.
- Designed a **BLE Mesh network** for secure node-to-node and many-to-one communication.
- Implemented **secure data encryption** over wireless networks, ensuring data security.
- Developed **power-optimized firmware** for battery-powered IoT devices, extending device life.
- Created **OTA firmware update mechanisms** over Bluetooth with public-private key security pairing.
- **Board bring-up and initial firmware testing** for new embedded hardware.

Embedded Engineer | Chiptest Engineering Ltd, Chennai **Jan 2019 – Jan 2020**

- Worked on Home and Industrial automation projects.
- Easily control the utilities and features of home appliances over the Internet.
- Built to work in both online and offline modes.

Executive – Manufacturing Services | NewGen Knowledge Works Pvt Ltd **Dec 2015 – Jan 2019**

EDUCATION

Bachelor of Electrical and Electronics Engineering
Thangavelu Engineering College, 2014

CERTIFICATIONS & TRAINING

- **Advanced Embedded Systems** – Vector India Institute, Chennai
- **IoT Hacking & Embedded Data Security** – Nullcon, Goa
- **Mastering RTOS & Embedded AI** – Udemy

PROJECT HIGHLIGHTS

- **Robotics Drivetrain Control:** Developed motor and encoder control software for robotics drivetrain systems using ROS and embedded MCU firmware, enabling reliable real-time motion control.
- **IoT Security Implementation:** Developed public-private key encryption and TLS security for MQTT-based IoT devices, enhancing security compliance.
- **Power Optimization for IoT:** Extended battery life by 30% through deep-sleep modes, adaptive duty cycling, and ultra-low-power firmware optimization.
- **Firmware Architecture:** Designed multi-threaded RTOS-based firmware with task prioritization and an event-driven state machine.
- **Automated Manufacturing Testing:** Created an automated testing framework for mass production, reducing testing time by 40%.
- **BLE Application Development:** Implemented custom BLE services and profiles for seamless device communication, improving BLE data throughput and connection stability.

TOOLS & TECHNOLOGIES

- **Development IDEs:** STM32CubeIDE, ESP-IDF, Keil, Segger Embedded Studio, VS Code
- **Debugging & Analysis:** JTAG, SWD, GDB, OpenOCD, Wireshark
- **CI/CD & DevOps:** GitLab CI/CD, Bitbucket
- **Embedded Security:** Secure Boot, Cryptographic Libraries, Key Management Systems
- **Robotics & Build Systems:** ROS, Catkin, Linux, STM32-Linux Build System

KEY ACHIEVEMENTS

- Took ownership as **component owner** for robotics embedded software at Rapyuta Robotics, driving feature development and cross-functional delivery.
- Developed and deployed multiple IoT products, scaling to **10,000+ devices** in production.
- Designed a **secure boot mechanism**, preventing unauthorized firmware access and improving device security.
- Reduced power consumption by **30%** through software-controlled dynamic power management.
- Created an **automated provisioning system** for IoT devices, reducing manual setup time by 50%.

I hereby declare that the above written particulars are true to the best of my knowledge and belief.